



Power Wedge Improvement Adapter Retention System GET 7495

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

**MARC
Management**

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July 2024
1.10-240709-1

1.0 Introduction

In the introduction of GET 7495 period 2015, 2016, 2017, 2018, 2019 no events were observed in the adapter hammerless system. In the year 2022 we present loose adapter wedge events and evaluate the need to implement a retention cap with a clamping screw.

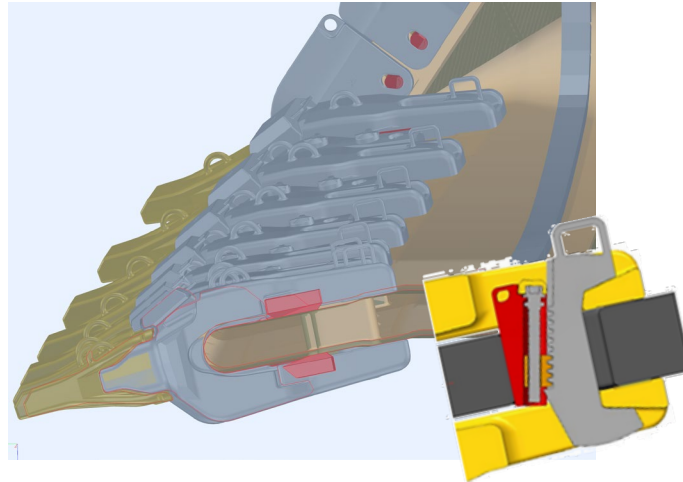


Fig 1. Review of the system in court

The 7495 machine is of high production and rarely do we have times of opportunity to be able to intervene in the equipment and perform adjustment checks on the 3983032-power wedge that allow us to maintain reliability in the System.

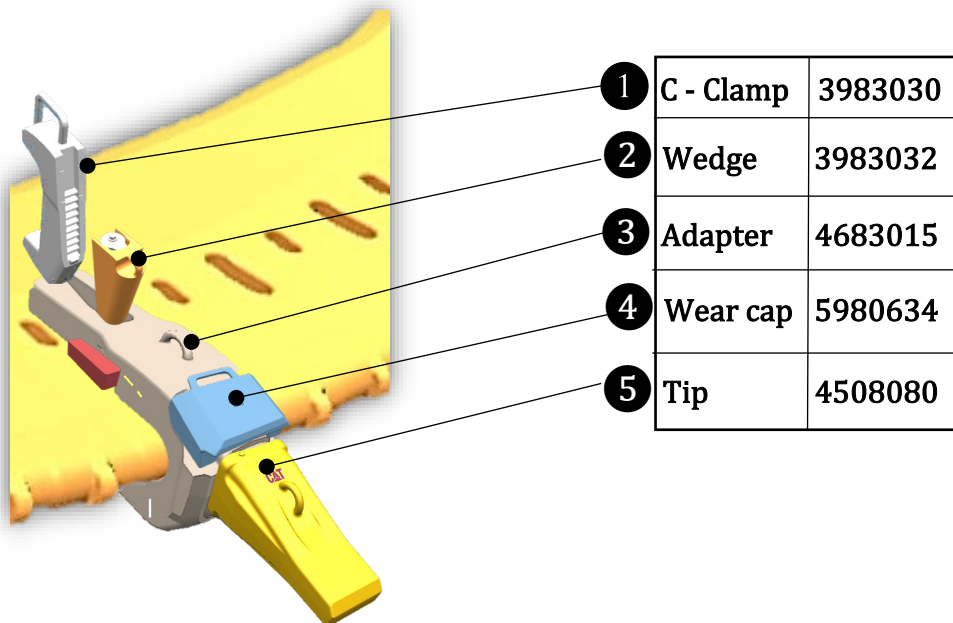


Fig 2. System Parts

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Fig 3. Operation Equipment.

The inspections that are carried out is a few minutes to check the adjustment of the wedges and the cleaning of the pieces that contain accumulated dirt, the need to improve the design of the wedge is created to reduce the frequency of inspection of the wedge, this action will allow us to increase the reliability of the product.

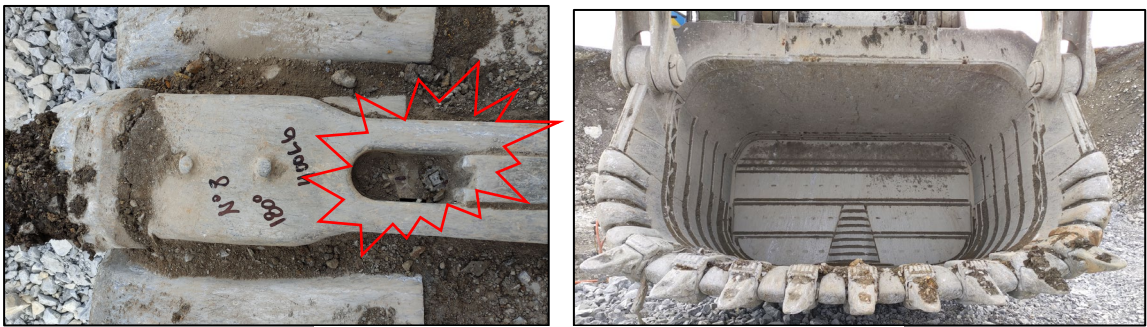


Fig 4. Dipper condition for inspection

After the events caused by the insurance, the following implementation was determined where the retention cover was redesigned, and a fastening screw was implemented.



Fig 5. Power wedge with improved retention cap

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2.0 Best Practice Description

To guarantee the permanence of the wedge in the adapter and avoid possible misalignment of the wedge, an improvement is made to the cover and a screw is added.

- The design of the wedge screw changed to be able to place a cover that contains a hexagonal head screw to hold the cover.

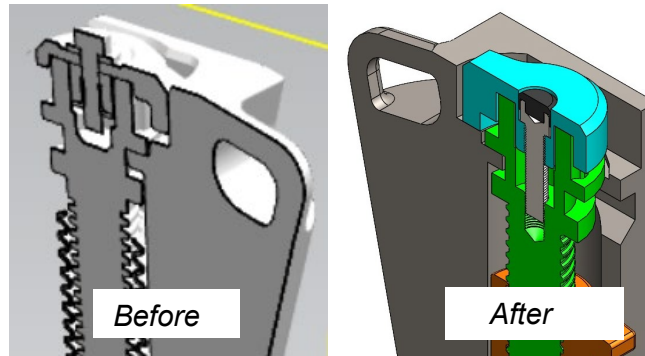


Fig 6. Parts of the new wedge

3.0 Implementation Steps

- A new cover with greater depth for the head of the screw was manufactured and drilled in the center of the body as shown in the following image of the power wedge to condition a hexagonal head bolt.

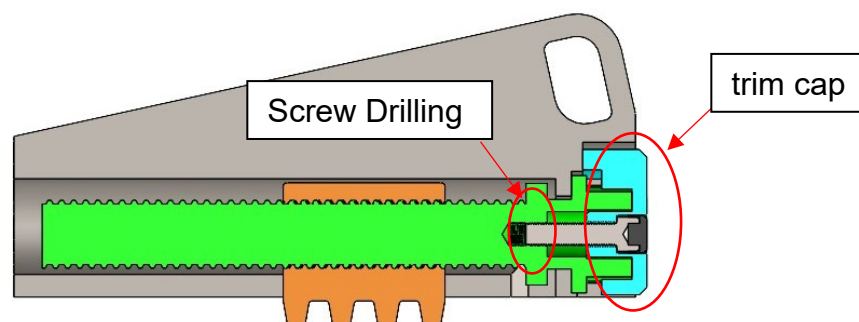
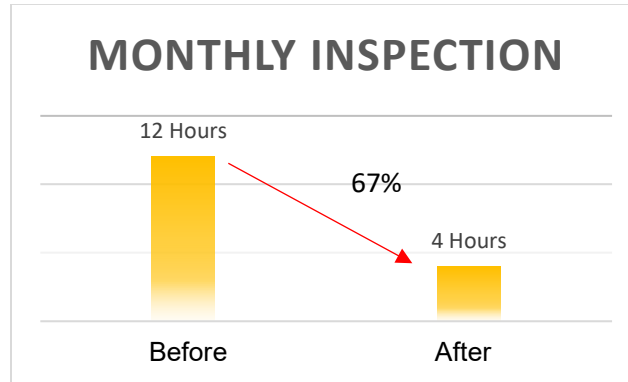


Fig 7. Power wedge side view

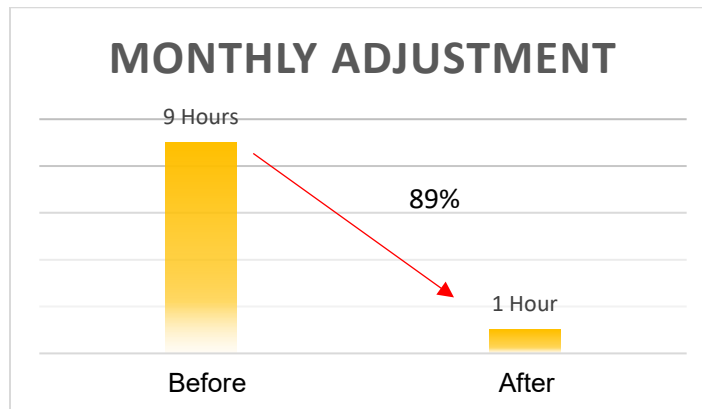
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4.0 Benefits

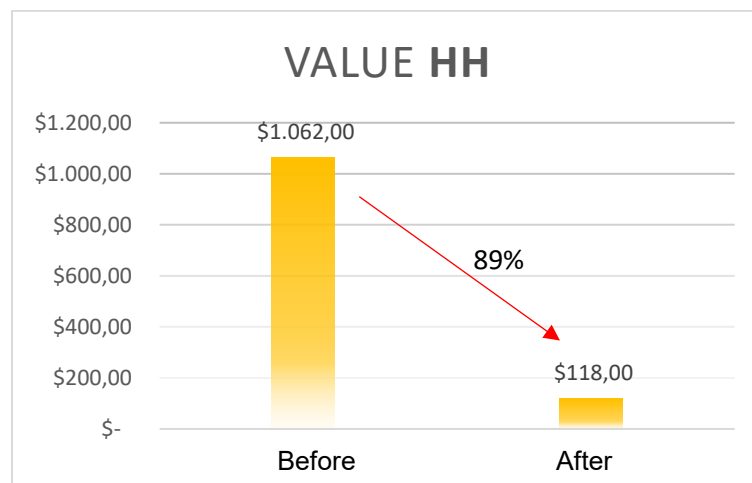
- Reduction of torque inspection time in wedge 3983032, reduced by 67% due to the anti-rotation operation.



- The power wedge readjustment time is reduced by 89% due 15 days interval.



- Reduction in HH value.



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- Reduction in shovel stop time.



- The following table indicates the time that the system lasted, without adjustments after the settlement of the system.

OPERATION	1st and 2nd ADJUSTMENT	ACCUMULATED HOURS WITHOUT ADJUSTMENT
QUELLAVECO	Correct	4845 Hours
ANTAPACCAY	Correct	1307 Hours



Fig 8. Power wedge ADJUSTMENT

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Fig 9. Power wedge ADJUSTMENT

5.0 Resources Required

- ❖ Design plans for manufacturing (see appendix).
- ❖ Improved hammerless safety arming.
- ❖ Manufacture of protective covers.

6.0 Supporting Attachments / References

- Data and monitoring of wedge settings.

Before using the improved wedge, the racket had 2 interventions between each preventive maintenance, now the insurance is adjusted after 24 hours, then 200 hours and every 500 hours.

Inspection Date	Inspection Hours	QUELLAVECO POSICIÓN										
		Status	Pos RH	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos LH
12/11/2021	0	Task	Installation	Installation	Installation	Installation	Installation	Installation	Installation	Installation	Installation	Installation
		Turn	360°	360°	360°	360°	360°	360°	360°	360°	360°	360°
1/12/2021	456	Task	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT
		Turn	360°	360°	360°	360°	360°	360°	360°	360°	360°	360°
7/01/2022	1213	Task	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT
		Turn	360°	360°	360°	360°	360°	360°	360°	360°	360°	360°
9/02/2022	1686	Task	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT
		Turn	360°	360°	360°	360°	360°	360°	360°	360°	360°	360°
24/03/2022	2563	Task	ADJUSTMENT	OK	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	OK	OK
		Turn	360°	360°	360°	360°	360°	360°	360°	360°	360°	360°
28/04/2022	3559	Task	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT
		Turn	360°	360°	360°	360°	360°	360°	360°	360°	360°	360°
4/05/2022	3592	Task	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT
		Turn	360°	360°	360°	360°	360°	360°	360°	360°	360°	360°
9/06/2022	4259	Task	OK	OK	OK	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	ADJUSTMENT	OK	OK	OK
		Turn	360°	360°	360°	360°	360°	360°	360°	360°	360°	360°
2/07/2022	4845	Task	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
		Turn	360°	360°	360°	360°	360°	360°	360°	360°	360°	360°

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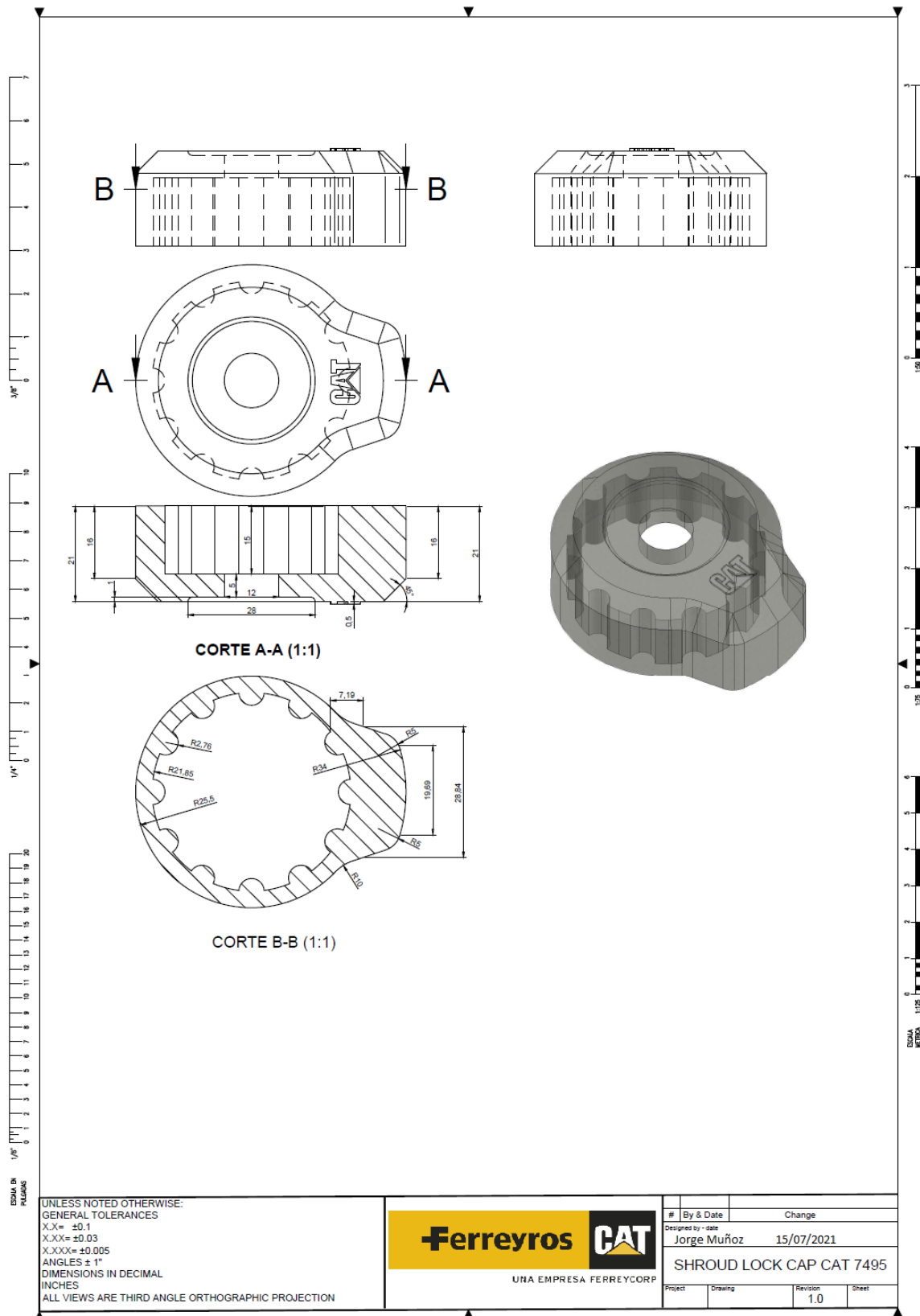


Fig 11. Wedge 3983032 Design Drawings 2 (drawing in appendix)

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7.0 Related Best Practices

8.0 Acknowledgements

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